



Cashew apple byproduct: Gastroprotective effects of standardized extract

Gisele Goulart da Silva^{a,b}, Lucia Elaine de Oliveira Braga^{a,b}, Ellen Cristina Souza de Oliveira^a, Sirlene Valério Tinti^a, João Ernesto de Carvalho^c, Josy Goldoni Lazarini^b, Pedro Luiz Rosalen^{b,d}, Ana Paula Dionísio^e, Ana Lucia Tasca Gois Ruiz^{c,*}

^a Chemical, Biological and Agricultural Pluridisciplinary Research Center, University of Campinas, CPQBA/UNICAMP, 13148-218, Paulínia, SP, Brazil

^b Postgraduate Program in Dentistry, Piracicaba Dental School, University of Campinas, FOP/UNICAMP, 13414-903, Piracicaba, SP, Brazil

^c Faculty of Pharmaceutical Sciences, University of Campinas, FCF/UNICAMP, 13083-871, Campinas, SP, Brazil

^d Biological Sciences Graduate Program, Federal University of Alfenas – UNIFAL-MG, 37130-001, Alfenas, MG, Brazil

^e Embrapa Agroindústria Tropical, 60511-110, Fortaleza, CE, Brazil

ARTICLE INFO

Keywords:

Anacardium occidentale L.
Cashew apple pomace
Carotenoids
Anacardic acids
Gastric ulcer

ABSTRACT

Ethnopharmacological relevance: The incidence of gastric mucosa lesions in the adult population has increased mainly due to the continued use of nonsteroidal anti-inflammatory drugs (NSAIDs). The cashew (*Anacardium occidentale* L.) is a tropical tree, cultivated in several countries, whose barks, leaves and pseudofruit (cashew apple) are popularly used in traditional medicine for the treatment of many diseases, including gastric ulcer.

Aim: Our study evaluated the potential gastroprotective effect of the carotenoid and anacardic acids-enriched aqueous extract (CAE), prepared from cashew apple pomace, in the dose-repeated acetylsalicylic acid (ASA)-induced gastric lesions model in rats.

Material and methods: After randomly distribution into five group (G1 – G5, n = 8 animals/group), male Wistar rats were daily treated with ASA solution (200 mg/kg, 5 ml/kg, G2 - G5) or potable water (Satellite group, G1) during 14 days. From 8th to 14th experimental day, rats in G3 - G5 groups were orally treated with CAE (50, 100 and 500 mg/kg, 5 ml/kg, respectively). Body weight was measured on 0, 7th and 14th day. On the 14th experimental day, all surviving animals were euthanized for macroscopic evaluation of the inner organs and stomach removal. After weighting, each stomach was properly prepared for biochemical analysis [myeloperoxidase activity (MPO), reduced glutathione analysis (GSH), IL-1 β , CXCL2/MIP-2, TNF- α and IL-10 levels].

Results: At the most efficient dose (100 mg/kg, p.o.), CAE-treated animals showed a slight improvement in the macroscopic aspect of gastric mucosa associated with significant (p < 0.05) reduced levels of IL-1 β , CXCL2/MIP-2, and MPO activity besides increased levels of GSH (partially), and IL-10 in stomach tissues.

Conclusions: The present study demonstrated that the carotenoid and anacardic acids-enriched extract obtained from cashew apple pomace is a promising raw material for the development of herbal medicine and/or functional food supplements for the adjuvant treatment of NSAIDs-induced gastric ulcers.

1. Introduction

Affecting a large part of the population, one of the main causes of gastric mucosa lesions is the continuous use of nonsteroidal anti-inflammatory drugs (NSAIDs) (Garbayo et al., 2017; Bjarnason et al., 2018). Among NSAIDs, acetylsalicylic acid (ASA) has been strongly correlated with the incidence of gastric lesions mainly due to its use as

an antiplatelet agent in the treatment of cardiovascular diseases (Huang et al., 2011; Bjarnason et al., 2018). As a nonspecific inhibitor of cyclooxygenases, ASA promotes inhibition of both inflammatory prostaglandins (PG) and those related to gastric cytoprotection. Blockage of cytoprotective PG synthesis leads to decreased blood flow besides mucus and bicarbonate secretion in the gastric mucosa, leading to increased incidence of gastric lesions (Huang et al., 2011; Takeuchi and Amagase, 2018). The strategies for control this problem involves, in general, both

* Corresponding author. Faculty of Pharmaceutical Sciences, University of Campinas, Cândido Portinari street, 200 Cidade Universitária Zeferino Vaz, 13081-970, Campinas, SP, Brazil.

E-mail addresses: giselegoulart87@gmail.com (G. Goulart da Silva), naine17@hotmail.com (L.E. de Oliveira Braga), ellen.oliveira87@yahoo.com.br (E.C. Souza de Oliveira), sirlenetinti@gmail.com (S. Valério Tinti), carvalho@fcp.unicamp.br (J.E. de Carvalho), Josy662@hotmail.com (J. Goldoni Lazarini), rosalen@fop.unicamp.br (P.L. Rosalen), ana.dionisio@embrapa.br (A.P. Dionísio), ana.ruiz@fcp.unicamp.br (A.L. Tasca Gois Ruiz).

<https://doi.org/10.1016/j.jep.2020.113744>

Received 28 September 2020; Received in revised form 4 December 2020; Accepted 21 December 2020

Available online 25 December 2020

0378-8741/© 2020 Elsevier B.V. This article is made available under the Elsevier license (<http://www.elsevier.com/open-access/userlicense/1.0/>).

Abbreviations

NSAIDs	Non-steroidal anti-inflammatory drugs
ASA	acetylsalicylic acid
PG	prostaglandin
GIT	gastrointestinal tract
CAE	Carotenoid and anacardic acids-enriched extract
PBS	phosphate-buffered saline
TCA	trichloroacetic acid
GSH	reduced glutathione
MPO	myeloperoxidase
HTAB	hexadecyltrimethylammonium bromide
H ₂ O ₂	hydrogen peroxide
NADPH	β -nicotinamide adenine dinucleotide 2'-phosphate reduced tetrasodium salt hydrate
DTNB	5,5'-dithiobis 2-nitrobenzoic acid

GSSG	oxidized glutathione
TNF- α	tumor necrosis factor α
IL-1 β	Interleukin 1- β
IL-10	Interleukin -10
CXCL-MIP-2	chemokine macrophage inflammatory protein 2
HPLC-DAD-MS	high performance liquid chromatography coupled to diode array detection and electrospray ionization tandem mass spectrometry
UPLC-ESI-QToF-MS	ultra high performance liquid chromatography coupled with spectrometry ion flight mass quadrupole electrospray
PGE ₂	prostaglandin E ₂
HOCl	hypochlorous acid
ROS	reactive oxygen species
COX	cyclooxygenase
HATs	histone acetyltransferases

the search for new NSAID with lower incidence of gastrointestinal side effects and the development of adjuvant gastroprotective drugs (Sen et al., 2009; Chi et al., 2018). In this context, herbal medicines traditionally used for the treatment of gastrointestinal problems, highlighting *Anacardium occidentale* L. (Lim, 2012; Carvalho et al., 2015), represent an important source of gastroprotective drugs.

Anacardium occidentale (Anacardiaceae family) is a tropical tree native in Colômbia, Venezuela, Guyana, and the North and Northeast of Brazil and cultivated in Central America, India, and several countries in East Africa (Santos et al., 2007; Salehi et al., 2019). In traditional medicine, leaves, fruits (cashew apple) and bark stem of *A. occidentale* are used in different preparations for the treatment of cutaneous, oral and gastric wounds and ulcers, dyspepsia, diarrhea, diabetes, and parasitosis (Lim, 2012; Da Silveira Vasconcelos et al., 2015; Salehi et al., 2020). Some pharmacological studies have demonstrated the anti-ulcerative effect of both bark stem extracts and cashew juice attributing this effect partially to the anacardic acids present in these preparations (Morais et al., 2010; Sharifi-Rad et al., 2018).

In addition, *A. occidentale* is an important income source in many countries through the marketing of the cashew nuts (true fruit), juices, beverages, jams, syrups, ice cream and sauces prepared from the cashew apple (pseudofruit) together with cashew gum (used as excipient) and wood (Akinwale, 2000; Oliveira, 2008; Ganesh et al., 2015). The cashew apple processing for juice production results in a great amount of residual fibers (pomaces) that were traditionally used as animal feed (Rocha et al., 2011; Das and Arora, 2017). However, many times discarded in nature generating several environmental damages (Esparza et al., 2020), these pomaces can be an interesting source of compounds with biological activities (Das and Arora, 2017). Studies conducted by Brazilian researchers from Embrapa (CE, Brazil) in partnership with CIRAD/France allowed the development of one intense-yellowish-colored aqueous extract (CAE) from cashew pomaces. The chemical analysis identified the presence of at least eleven different carotenoids and three different classes of anacardic acids in CAE (Abreu et al., 2013a, 2013b; Dionísio et al., 2018; Sousa et al., 2020).

To evaluate the efficiency of new gastroprotective agents, some NSAIDs-induced gastric injury animal models have been developed (Kauffman, 1989; Perini et al., 2004; Adinortey et al., 2013; Simões et al., 2019). In these experimental models, the evaluation of sample influence on the NSAIDs-induced gastric lesions caused can be accompanied by several parameters, including myeloperoxidase activity, reduced glutathione, and pro- and anti-inflammatory cytokines and chemokines dosages (Mosser and Zhang, 2008; Parker and Winterbourn, 2013; Bjarnason et al., 2018).

Considering the traditional use of anacardic acids-enriched extracts prepared from different parts of *A. occidentale* besides the presence of

these substances together with some carotenoids in the intense-yellowish-colored aqueous extract (CAE) prepared from cashew pomaces, the present study evaluated the possible antiulcerogenic activity of CAE in the dose-repeated acetylsalicylic acid (ASA)-induced gastric lesions model in rats aiming the future use of CAE as a raw material for the development of phytotherapeutic drugs and/or functional food supplements.

2. Material and methods

2.1. Plant material

The CAE used in all experiments was obtained from the cashew apple (*Anacardium occidentale*, variety CCP-76) pomaces (1:1 w/w, pomaces: water) through pressing and filtration techniques followed by lyophilization as described by Abreu et al. (2013a and 2013b) and Sousa et al. (2020). Chemical characterization (HPLC-DAD-MS, UPLC-QToF-ESI and colorimetric methods) allowed the identification of carotenoids (total carotenoids $\approx$ 24 mg/100 g), being β -cryptoxanthin and auroxanthin isomers the major compounds ($\approx$ 50% of total carotenoids) besides monoene, diene and triene anacardic acids ($\approx$ 443 mg/100g of total anacardic acids) (Abreu et al., 2013a; Dionísio et al., 2018; Sousa et al., 2020). After been chemically characterized by one of the authors (A.P. D., Embrapa - Fortaleza) as already have been described in literature (Abreu et al., 2013a; Dionísio et al., 2018; Sousa et al., 2020), the CAE was packaged in amber glass bottles and kept at -80 °C until before pharmacological evaluations.

2.2. Animals

Specific pathogen-free male HanUnib: WH (Wistar) rats (*Rattus norvegicus*, 4 weeks old, forty animals) were obtained from the Multi-disciplinary Center for Biological Research in Animal Science (CEMIB/UNICAMP) after approval by the Animal Ethics Committee (CEUA/UNICAMP), protocol number 4345-1. Animal care, as well as experimental protocols, were performed in accordance with the principles and guidelines adopted by the Brazilian National Council for Animal Experiment Control (CONCEA) e European Union Regarding Animal Experimentation (Directive of the European Council 86/609/EC).

The animal was housed (5 rats/cage) throughout the period in polyethylene boxes (matte white) with disinfected softwood beddings, at room temperature of 22 ± 2 °C; relative humidity of $50 \pm 20\%$ and light/dark cycle of 12 h/12 h. All rats received commercial pelleted feed Nuvilab® and potable water *ad libitum*. For the experiments, all animals reached 12 weeks of life (body weight 300–450 g). Euthanasia was done by deepening anesthesia with a mixture of ketamine (Dopalen

injectable® 300 mg/kg) and xylazine (Anasedan injectable® 30 mg/kg) followed by cervical dislocation.

2.3. Chemical

Ketamine (Dopalen injectable®), xylazine (Anasedan injectable®), tween 80 (Synth), carboxymethyl cellulose (CMC) (Sigma Aldrich C9481), acetylsalicylic acid (ASA) (sigma Aldrich A5376), hexadecyltrimethylammonium bromide (HTAB) (Sigma Aldrich H5882), *o*-dianisidine dihydrochloride (Sigma Aldrich D3252), hydrogen peroxide (Synth), horseradish peroxidase (Sigma Aldrich P8250), trichloroacetic acid (TCA) (Sigma Aldrich T4885), reduced L-glutathione (Sigma Aldrich G-4251), β -nicotinamide adenine dinucleotide 2'-phosphate reduced tetrasodium salt hydrate (NADPH) (Sigma Aldrich N-7505), 5,5'-dithiobis (2-nitrobenzoic acid) (DTNB) (Sigma Aldrich D-8130), buffer RIPA (R0278), IL-1 β mouse, IL-10 mouse, CXCL2-MIP-2 mouse, and TNF- α mouse were obtained by R&D Systems, Inc (Minneapolis, Minnesota, USA).

2.4. Sample preparation and treatment techniques

CAE was mixed with Tween 80 (Sigma Aldrich, final concentration = 0.2%) followed by dilution in 0.2 M phosphate buffer solution (pH 7) prepared in 0.9% NaCl solution (PBS). The PBS plus Tween 80 0.2% constituted the vehicle. Due to low water solubility of acetylsalicylic acid (ASA), first carboxymethylcellulose (CMC, 0.2% w/v) was slowly suspended in hot distilled water, under mechanical agitation. Then, finely ground ASA was mixed resulting in a homogeneous suspension. All oral treatments were done by gavage using a flexible orogastric catheter ($\varnothing$ 2 mm, 5 cm length).

2.5. Repeated-dose ASA-induced ulcer model

This experimental protocol was designed based on those described by Jin et al. (1999) and Pimple et al. (2012) with few adaptations. Forty male Wistar rats were weighed and randomly distributed into five groups ($n = 8$ animals/group, G1 - G5). Animals in G2 - G5 groups were orally treated with ASA solution (200 mg/kg, 5 ml/kg) during 14 days, while the animals of G1 group remained untreated (satellite group). On the 8th experimental day, 30 min after ASA treatment, the animals in the G3 - G5 groups were orally treated with CAE (50, 100 and 500 mg/kg, 5 ml/kg) until the 14th experimental day. The treatment sequence was from the highest to the lowest CAE dose. Body weight was measured on 0 (baseline), 7th, and 14th experimental days. On the 14th day, all surviving animals were euthanized for macroscopic evaluation of the inner organs and stomach removal. On an icy surface, each stomach was opened through the small curvature, washed with PBS, and cut into four fragments. Three fragments were frozen and stored at -80 °C for myeloperoxidase activity (MPO), cytokines, and chemokines analysis. One fragment of each stomach was weighed and stored in trichloroacetic acid solution (TCA 5%, 1 ml/fragment) at -80 °C for reduced glutathione (GSH) analysis. All fragments were stored individually.

2.5.1. Myeloperoxidase (MPO) activity assay

The MPO activity assay followed the protocol described by Krawisz et al. (1984). After thawing and weighing, one fragment of each stomach was homogenized by sonication with hexadecyltrimethylammonium bromide solution (HTAB 0.5% at PBS 50 mM pH 6.0, 1:20 w/v). After three cycles of freezing and thawing, the homogenates were centrifuged (7000 g, 10 min, 4 °C). Two aliquots (50 μ L/well) of each supernatant were transferred to a 96-well plate and mixed with reactive solution [150 μ L/well; dihydrodrate of *o*-dianisidine (0.167 mg/ml) and H₂O₂ (0.0005%) in PBS 50 mM, pH 6.0]. Absorbance at 450 nm was recorded using the VersaMax microplate reader (Molecular Devices) at 37 °C. Using a calibration curve of horseradish peroxidase (0.15–78 U), the results were expressed in U/g of tissue.

2.5.2. Determination of glutathione

The glutathione determination assay followed the protocol described by Anderson et al. (1985). After thawing, the TCA volume was adjusted to 1:10 w/v in each fragment stored in TCA before sonication and centrifugation (2000 g, 10 min, 4 °C). Two aliquots (20 μ L/well) of each supernatant were transferred to a 96-well plate and mixed with NADPH (140 μ L/well, 0.3 mM), PBS (5 μ L/well, 50 mM pH 6.0) and DTNB (20 μ L/well, 6 mM) solutions. After incubation for 5 min at 30 °C, the GSSG reductase solution (4 IU/ml, in PBS, 15 μ L/well) was added. Absorbance at 412 nm was recorded using the VersaMax microplate reader (Molecular Devices). Using a reduced glutathione calibration curve (0.15–10 nmol/100 μ L), the results were expressed in nmol/g of tissue.

2.5.3. Cytokines (TNF- α , IL-1 β , IL-10) and chemokine (CXCL2/MIP-2) levels

After thawing, one fragment of each stomach was sonicated in extraction buffer [RIPA (1:3 w/v) plus protease inhibitor (1: 0.03 w/v) and PBS (50 mM pH 6.0) q.o. p 700 μ L/sample] and centrifuged (2000 g, 10 min, 4 °C). The protein level was determined by the Bradford method. The tissue levels of TNF- α , IL-1 β , IL-10 and CXCL2/MIP-2 were determined by ELISA immunoassay on microplate reader. The protocol was protocols supplied by the manufacturer's instructions (R&D Systems®) The Proteins quantity were normalized and results expressed in pg/mg of protein.

2.6. Statistical analysis

The results were expressed as mean $\pm$ standard error or as boxplot graphics. The difference between the means was determined by the analysis of variance ANOVA (One-way or Two-way, depending on the parameter) followed by the Bonferroni test. The analyses were performed in the GraphPadPrism software version 5.0 (GraphPad Software). A significant difference was those with $p < 0.05$.

3. Results

In the ASA-induced ulcer model, animals were weekly monitored for the body weight evolution. After seven days of ASA treatment (200 mg/kg), all animals in G2 - G5 groups showed less weight gain in comparison to those from the Satellite group (G1). On the 14th experimental day, the tendency of lower weight gain for ASA-treated animals compared to those in the satellite group was maintained independently of CAE-treatment (Table 1). During the macroscopic analysis, the ASA-treated stomachs showed petechiae presence besides generalized redness in the gastric mucosa in comparison to the stomachs of animals in the Satellite group (data not shown).

At the end of the experiment, the stomachs of ASA-treated animals (G2) showed a significant ($p < 0.05$) increase in MPO activity followed by a depletion of GSH level in comparison to animals in the Satellite group (G1) (Fig. 1). The concomitant CAE treatment at 100 and 500 mg/kg, p.o. (G4 and G5, respectively), started at the 8th experimental day, reverted the MPO activity to similar activity as observed to the animals in the Satellite group (G1). More, CAE-treated animals at G4 and G5 also showed partial increase (statistically similar both to G1 and G2 groups) in the GSH level except by at 50 mg/kg dose level (Fig. 1).

As CAE treatment at 50 mg/kg was not able to reverse ASA-induced alterations in MPO activity and GSH level (Fig. 1), the IL-1 β , IL10, TNF- α , and CXCL2/MIP-2 levels were analyzed in the stomachs of CAE-treated animals at 100 and 500 mg/kg (G4 and G5, respectively) in comparison to untreated (G1) and ASA-treated (G2) animals. None significant alterations on TNF- α , CXCL-MIP2, IL-1 β , and IL10 levels were observed between ASA-treated (G2) stomachs in comparison to those from untreated animals (G1) (Fig. 2). In comparison to ASA-treated animals (G2), CAE treatment at 100 mg/kg (G4) resulted in significant ($p < 0.05$) increase on IL-10 and TNF- α levels (Fig. 2A and D) besides a significant ($p < 0.05$) decrease in IL-1 β and CXCL-MIP2 levels (Fig. 2B

Table 1

Body weight evolution of Wistar rats in the repeated-dose ASA-induced ulcer model.

Group	weeks				
	0	1	2	3	4
	Basal Weight	Weight t	Δ	Weight t	Δ
G1	372.9 ± 15.1	439.1 ± 13.6	66.2 ± 18.7	455.8 ± 6.9	82.93 ± 16.4
G2	425.3 ± 15.3	417.6 ± 10.0	-7.7 ± 18.0***	462.0 ± 14.5	36.67 ± 3.7**
G3	454.4 ± 12.8	468.4 ± 13.2	14.0 ± 3.4**	491.0 ± 15.1	36.59 ± 3.7**
G4	467.5 ± 9.0	489.0 ± 10.0	21.5 ± 1.8*	514.9 ± 10.8	47.38 ± 4.5*
G5	429.8 ± 6.0	447.7 ± 5.9	17.9 ± 3.2**	468.8 ± 7.9	39.0 ± 6.7**

Results expressed as mean ± standard error; animals: male Wistar rats (12 week-old, n = 8 animals/group). Δ = body weight on 7th or 14th experimental day – basal body weight. Treatments: untreated animals (G1, Satellite group); acetylsalicylic acid (ASA, 200 mg/kg in 0.2% CMC solution in PBS, 5 ml/kg, G2 - G5, during 14 days); carotenoid and anacardic acids-enriched extract obtained from cashew pomace (CAE, 50, 100 and 500 mg/kg, G3, G4, and G5, respectively, diluted in PBS plus Tween 80 0.2%, from the 8th to the 14th experimental day). Statistical analysis by Two-way ANOVA followed by Bonferroni's test (*p < 0.05; **p < 0.01; ***p < 0.001, related to the G1 group) using the GraphPad Prism 5.0 software.

and C) while at 500 mg/kg, CAE treatment just promoted a significant (p < 0.05) reduction on IL-1β level (Fig. 2B).

Results expressed as boxplot graphics. Male Wistar rats (12 week-old, n = 8 animals/group). Treatments: untreated animals (G1, Satellite group); acetylsalicylic acid (ASA, 200 mg/kg in 0.2% CMC solution in PBS, 5 ml/kg, G2 - G5, during 14 days); carotenoid and anacardic acids-enriched extract obtained from cashew pomace (CAE, 50, 100 and 500 mg/kg, G3, G4, and G5, respectively, diluted in PBS plus Tween 80 0.2%, from the 8th to the 14th experimental day). Statistical analysis: one-way ANOVA followed by the Bonferroni test (similar letters indicate non-significant differences among results) using GraphPad Prism5.

Results of TNF-α (A, pg/ml), CXCL-MIP2 (B, pg/ml), IL-1β (C, pg/ml), and IL-10 (D, pg/ml) levels expressed as boxplot graphics. Male Wistar rats (12 week-old, n = 8 animals/group). Treatments: untreated animals (G1, Satellite group); acetylsalicylic acid (ASA, 200 mg/kg in 0.2% CMC solution in PBS, 5 ml/kg, G2 - G5, for 14 days); carotenoid and anacardic acids-enriched extract obtained from cashew pomace (CAE, 50, 100 and 500 mg/kg, G3, G4, and G5, respectively, diluted in PBS plus Tween 80 0.2%, from the 8th to the 14th experimental day). Statistical analysis: one-way ANOVA followed by the Bonferroni test (similar letters indicate non-significant differences among results) using GraphPad Prism5.

4. Discussion

The present study showed the potential beneficial effect of the carotenoid and anacardic acids-enriched extract (CAE) obtained from the cashew apple pomace, a byproduct of cashew juice production, on dose-repeated NSAID-induced gastric lesions. Thus, from a byproduct of the cashew juice industry and using environmental-friendly techniques, such as pressing and filtration techniques (Abreu et al., 2013a, 2013b), it was prepared one standardized product that could be used as raw material for the development of herbal medicines and/or functional food supplement for the treatment of gastrointestinal disorders.

Chemical analysis of CAE demonstrated the presence of at least two classes of natural products. The total carotenoids content was established as 24.46 ± 0.15 mg/100 g (Sousa et al., 2020). By HPLC-DAD-MS, eleven carotenoids were identified being auroxanthin isomers and β-cryptoxanthin the major components (Abreu et al., 2013a). More,

UPLC-ESI-QToF-MS and HPLC-DAD methods allowed to infer the presence of triene (15:3, 108.50 ± 0.71 mg/100 g CAE), diene (15:2, 124.00 ± 0.00 mg/100 g CAE) and monoene (15:1, 210.50 ± 6.36 mg/100 g CAE) anacardic acids in CAE (Dionísio et al., 2018; Sousa et al., 2020). Both carotenoids and anacardic acids have been described as anti-inflammatory and anti-carcinogenic agents probably due to their antioxidant activities (Kubo et al., 2006; Morais et al., 2010; Hemshelkar et al., 2012; Encarnação et al., 2016; Amengual, 2019; Sharifi-Rad et al., 2018; Salehi et al., 2020).

In this context, we evaluated CAE in the dose-repeated ASA-induced gastric lesions model. The experimental dose range was established based on previous toxicological study that determined 500 mg/kg as the maximum tolerable dose of CAE for dose-repeated treatment schedule (da Silva et al., 2020). The lower weight gain induced by ASA-repeated doses can be attributed to discomfort during food intake and/or changes in the ability to assimilate nutrients, both promoted by gastric lesions induced by ASA (Malfertheiner et al., 2009; Cryer and Mahaffey, 2014). Regardless of the administered dose, the CAE was not able to reverse this side effect of ASA.

Both by inhibiting PGE₂ synthesis and inducing mitochondrial oxidative stress in gastric mucosa, ASA can induce gastric ulcers. This initial tissue damage both induces neutrophils migration and production/secretion of pro-inflammatory cytokines initiating the inflammatory cascade (Musumba et al., 2009; Turner et al., 2014; Bindu et al., 2020). One of the first cytokines released is the tumor necrosis factor α (TNF-α), which is present in macrophages, monocytes and lymphocytes. TNF-α modulates several biological responses, such as inducing the expression and releasing of other cytokines and chemokines, resulting in increased adhesion molecules expression and neutrophil activation (Watanabe et al., 2004; Musumba et al., 2009; Amirshahrokhi and Khalili 2016; Bindu et al., 2020). TNF-α also acts on macrophages and monocytes by increasing the chemokine macrophage inflammatory protein 2 (CXCL2/MIP-2) secretion in the tissue. This chemokine, also known as CXC ligand, is released by epithelial cells in the tissue injury and is a chemo-attracting favoring the entry of neutrophils into the tissue (Miller and Krangel, 1992; Watanabe et al., 2004; Rajarathnam et al., 2019).

Since the neutrophil by diapedesis migrate into the injured tissue, these cells will release MPO that mediate antimicrobial and phagocytic actions leading to the release of reactive oxygen species (ROS). Thus, the MPO activity is considered an inflammatory process indicator (Klebanoff, 2005; Parker and Winterbourn, 2013). Moreover, ROS excess can lead to depletion of endogenous antioxidant, such as reduced glutathione (GSH). This situation (↑MPO and ↓GSH) facilitates the tissue damage (Bjarnason et al., 2018).

Under normal conditions (intact gastric mucosa), the interleukin 1β (IL-1β) release down regulate the gastric acid production by acting on enterochromaffin, parietal and G cells, besides increasing the PGE₂ synthesis. (Wallace et al., 1992; Nakamura et al., 1998; Watanabe et al., 2004; Wallace 2008). When a gastric mucosal damage occurs, TNF-α-induced IL-1β release by monocytes and lymphocytes has a pro-inflammatory action, leading to ROS increased levels corroborating to the endogenous antioxidant depletion (Watanabe et al., 2001, 2004; El-Benna et al., 2005). In turn, IL-10 is defined as the anti-inflammatory cytokine, being responsible for limiting and/or terminating inflammatory processes by inhibiting the synthesis and release of cytokines and chemokines such as TNF-α, IL-1β, CXCL2/MIP-2 and IL-10 through regulating transcription factors such as NF-KappaB (Berg et al., 2001; Moore et al., 2001; Mosser and Zhang, 2008).

In this sense, our results suggested that the gastroprotective effect of CAE, at 100 mg/kg (most efficient dose), involved increasing levels of IL-10 besides reduced levels of CXCL2/MIP-2 and IL-1β. By providing low chemotactic gradient (↓CXCL2/MIP-2), CAE promoted reduction on MPO activity. Both IL-1β reduced level and lower MPO activity corroborated to reducing the damage caused by ROS and consequently restoring GSH level. More, by restoring IL-1β reduced level, the gastric

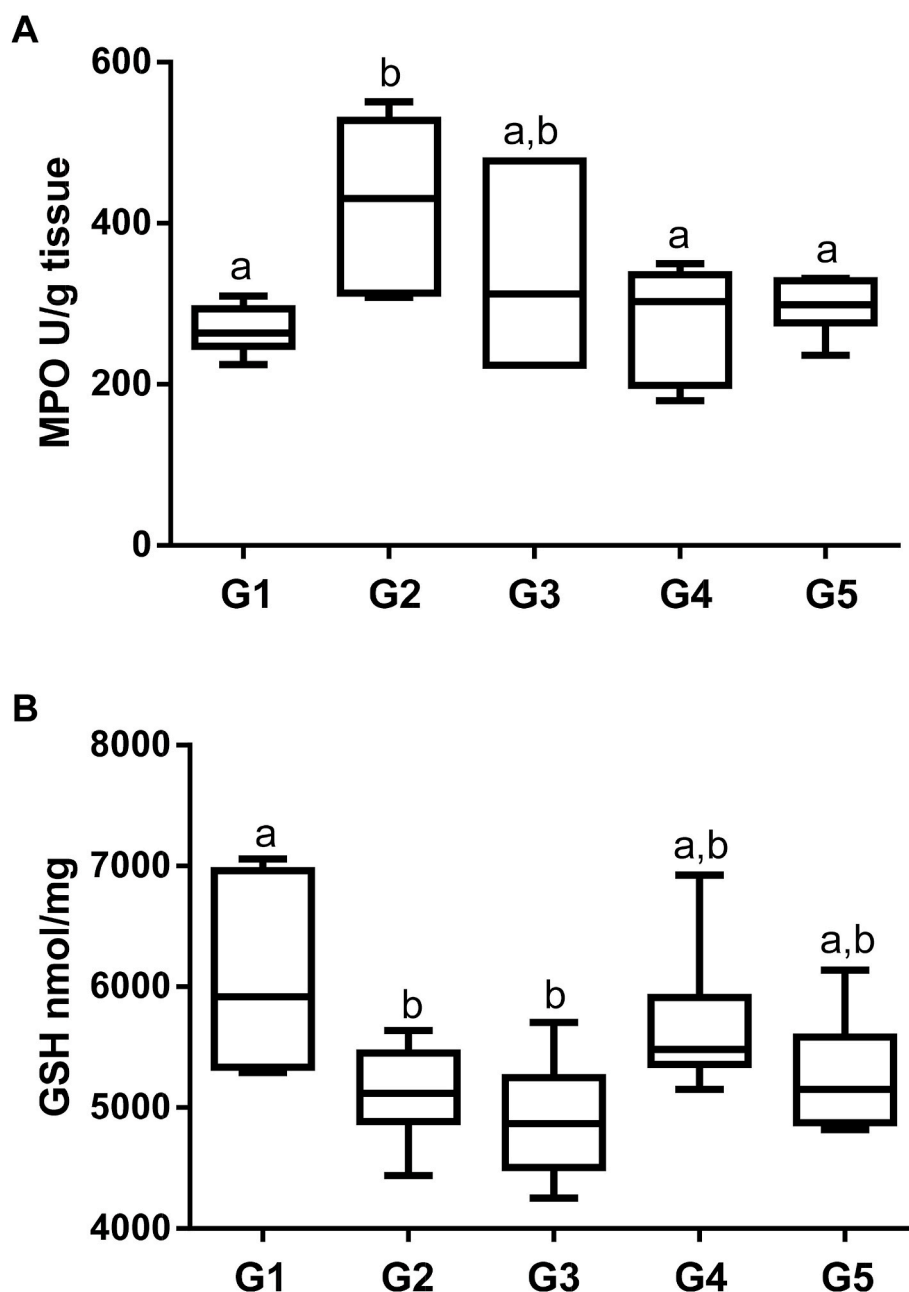


Fig. 1. Myeloperoxidase activity (A) and reduced glutathione level (B) in the stomachs of Wistar rats in the repeated-dose ASA-induced ulcer model.

acid production might be down regulated corroborating to slight improvement in macroscopic signs (petechiae and redness) observed in CAE-treated animals. This effect can be partly attributed to the carotenoids and anacardics acids identified in CAE.

Carotenoids are endogenous antioxidants acquired through vegetable ingestion (Fiedor and Burda, 2014; Milani et al., 2017). Epidemiological evidence has shown that the consumption of a carotenoid-rich diet is directly related to the lower incidence of some types of cancer, such as stomach (Persson et al., 2008) and colon (Kim et al., 2007) cancers, besides lower incidence of cardiovascular diseases, macular degenerations and cataract formation (Saini et al., 2015). These beneficial effects can be explained both by the scavenger ability and by the modulation of several cascades of pro-inflammatory signaling, including decreasing synthesis and release of cytokines, such as TNF- α and IL-1 β (Fiedor and Burda, 2014; Kaulmann and Bohn, 2014). As the carotenoid-mediated actions are dependent on the chemical structure of each substance (Britton, 1995), the effects observed in CAE-treated

animals may be a synergistic result of the different carotenoids already identified in CAE. Notably, β -cryptoxanthin (identified in CAE) daily consumption (plasma concentration $\approx 3 \mu\text{M}$) is related to a decreased risk of colon cancer among other pathologies (Burri, 2015; Burri et al., 2016; Lu et al., 2015).

Further, the anacardics acids are 6-alkylsalicylic acid derivatives also known as ginkgolic acids (van Beek and Montoro, 2009). More, the unsaturated (1–3 unsaturations) C15 side chain were the most common alkylsalicylic acid derivatives in cashew (Salehi et al., 2019). This class of natural products has been described as antioxidants, gastroprotective, anti-tumor, and antibacterial agents besides inhibitory effects against different enzymes such as acetylcholinesterase and tyrosine kinases (Hemshekhkar et al., 2012; Encarnação et al., 2016; Morais et al., 2017; Salehi et al., 2019). Both gastric protective activity and antitumor action seem to be related to the epigenetic regulation as histone acetyltransferases (HATs) inhibition that can modulate inflammatory cytokines, such as IL-1 β . In addition, in the gastric mucosa, anacardic acids

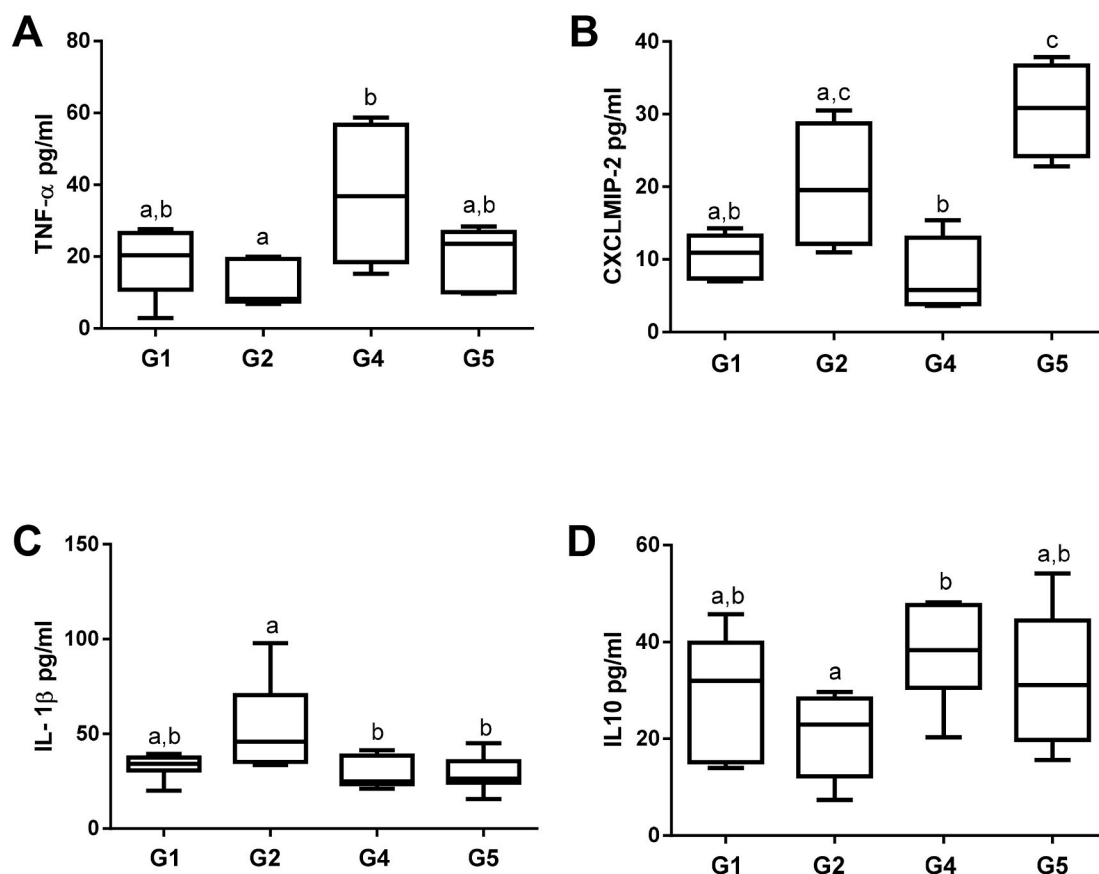


Fig. 2. Levels of IL-1 β , IL-10, TNF- α , and CXCLMIP-2 in the stomachs of Wistar rats in the repeated-dose ASA-induced ulcer model.

could restore to normality the nitric oxide levels and the superoxide dismutase and catalase activities (Sung et al., 2008; Morais et al., 2010; Hamad and Mubofu, 2015).

The dose-effect relationship observed for CAE might be explained by pharmacokinetic and pharmacodynamic issues. At 50 mg/kg, probably the active substances reached tissue concentrations below those necessary for the pharmacological effect on the gastric mucosa. In turn, increasing doses up to 500 mg/kg did not result in a proportional increase in the pharmacological effect of CAE suggesting the influence of the CAE formulation on the pharmacological effect. Regarding to the maximum volume (10 ml/kg) recommended for oral administration in rodents (Diehl et al., 2001), all dosages were prepared in almost similar volumes of vehicle, resulting in more concentrated solution at 500 mg/kg dose level than at 100 mg/kg. Due to the lipophilic characteristics (Gruszecki and Strzałka, 2005; Kaulmann and Bohn, 2014; Salehi et al., 2019), the emulsification degree of the carotenoids and anacardic acids might influence absorption through simple diffusion by enterocytes (Kiela and Ghishan, 2016). At 500 mg/kg, the higher CAE concentration might limit the bioactive compounds absorption and consequently the pharmacological effects.

Finally, based on the dose translation formula between species (Nair and Jacob 2016) and our results in rats, the estimated gastroprotective oral dose of CAE in humans (adults, 60 kg) for initial clinical studies was calculated as 16.2 mg/kg.

5. Conclusion

The present study showed for the first time that CAE has gastroprotective effect mediated by decreasing pro-inflammatory mediators (MPO, IL-1 β , and CXCL2/MIP-2) and increasing in anti-inflammatory and antioxidant mediators (GSH and IL-10) in the dose-repeated ASA-induced gastric lesions model in rats. Considering the huge amount of

cashew pomace generated in Brazil by cashew juice processing (around 300.000 of tons per year) (Araújo et al., 2018), our study contributes with one more alternative for a environmental-friendly destination of this cashew byproduct. This standardized extract (CAE) prepared from a byproduct of cashew juice processing can be used as raw material for the development of herbal medicine and/or a functional food supplement for the adjuvant treatment of NSAIDs-induced gastric ulcers.

Funding

This work was supported by the Coordination for the Improvement of Higher Education Personnel (CAPES, Financing Code 001) and the National Council for Scientific and Technological Development (CNPq, grant number 447191/2014–4 and grant number 429463/2018–9). The authors A.L.T.G.R., J.E.C and P.L.R. also grant CNPq for research productivity fellowship.

Author's contributions

Study conception and design: A. L. T. G. Ruiz, G. G. da Silva, A. P. Dionísio. Acquisition of data: A. P. Dionísio, G. G. da Silva, L. E. O. Braga, E. C. S. de Oliveira, S. V. Tinti, J. G. Lazarini. Analysis and interpretation of data: A. P. Dionísio, G. G. da Silva, L. E. O. Braga, E. C. S. de Oliveira, J. G. Lazarini. Drafting of manuscript: A. L. T. G. Ruiz, G. G. da Silva. Critical revision: A. L. T. G. Ruiz, J. E. de Carvalho, P. L. Rosalen, A. P. Dionísio, E. C. S. de Oliveira. Supervision and research resources supply: A. L. T. G. Ruiz, J. E. de Carvalho, P. L. Rosalen, A. P. Dionísio.

Declaration of competing interest

The authors declare no conflict of interest.

Acknowledgment

The authors thank the Chemical, Biological, and Agricultural Pluri-disciplinary Research Center (CPQBA/Unicamp) for the laboratory infrastructure.

References

- Abreu, F.P., Dornier, M., Dionisio, A.P., Carail, M., Caris-Veyrat, C., Dhuique-Mayer, C., 2013a. Cashew apple (*Anacardium occidentale* L.) extract from by-product of juice processing: a focus on carotenoids. *Food Chem.* 138 (1), 25–31. <https://doi.org/10.1016/j.foodchem.2012.10.028>.
- Abreu, F.A.P., Dornier, M., Pallet, D., Reynes, M., Vaillant, F., Furlani, F.C.T., 2013b. Method for the Concentration and Purification of the Extract Obtained from Cashew Pseudofruit Waste and Product with a High Carotenoid Content. *WO 2013/159167 A1*. 19/04/2013.
- Adinortey, M.B., Ansah, C., Galyuon, I., Nyarko, A., 2013. In vivo models used for evaluation of potential anti-gastrointestinal ulcer agents. *Ulcers*. Article ID 796405. <https://doi.org/10.1155/2013/796405>.
- Akinwale, T., 2000. Cashew apple juice: its use in fortifying the nutritional quality of some tropical fruits. *Eur. Food Res. Technol.* 211, 205–207. <https://doi.org/10.1007/s002170050024>.
- Amengual, J., 2019. Bioactive properties of carotenoids in human health. *Nutrients* 11 (10), 2388. <https://doi.org/10.3390/nu11102388>.
- Amirshahrokhi, K., Khalili, A.R., 2016. Gastroprotective effect of 2-mercaptoethane sulfonate against acute gastric mucosal damage induced by ethanol. *Int. Immunopharm.* 34, 183–188. <https://doi.org/10.1016/j.intimp.2016.03.006>.
- Anderson, M.E., 1985. Determination of glutathione and glutathione disulfide in biological samples. *Methods Enzymol.* 113, 548–555. [https://doi.org/10.1016/s0076-6879\(85\)13073-9](https://doi.org/10.1016/s0076-6879(85)13073-9).
- Araújo, W.F., Rocha, L.M., Araújo, I.M.S., de Paula, G.A., de Sousa, L.S., Folha, M.F., da Rocha Filho, L.B., Araújo, R.V., 2018. Sustentabilidade em agroindústrias: alternativas para evitar o desperdício de resíduos agroindustriais do pedúnculo de caju - uma revisão de literatura [Sustainability in agroindustries: alternatives to avoid waste of agro-industrial residue peduncle cashew - a literature review]. *Braz J of Develop* 4 (7), 4546–4569. Available at: <https://www.brazilianjournals.com/index.php/BRJD/article/view/714>.
- Berg, D.J., Zhang, J., Lauricella, D.M., Moore, S.A., 2001. IL-10 is a central regulator of cyclooxygenase-2 expression and prostaglandin production. *J. Immunol.* 166 (4), 2674–2680. <https://doi.org/10.4049/jimmunol.166.4.2674>.
- Bindu, S., Mazumder, S., Bandyopadhyay, U., 2020. Non-steroidal anti-inflammatory drugs (NSAIDs) and organ damage: a current perspective [published online ahead of print, 2020 Jul 10]. *Biochem. Pharmacol.* 180, 114147. <https://doi.org/10.1016/j.bcp.2020.114147>.
- Bjarnason, I., Scarpignato, C., Holmgren, E., Olszewski, M., Rainsford, K.D., Lanas, A., 2018. Mechanisms of damage to the gastrointestinal tract from nonsteroidal anti-inflammatory drugs. *Gastroenterology* 154 (3), 500–514. <https://doi.org/10.1053/j.gastro.2017.10.049>.
- Britton, G., 1995. Structure and properties of carotenoids in relation to function. *Faseb. J.* 9 (15), 1551–1558. PMID: 8529834.
- Burri, B.J., 2015. Beta-cryptoxanthin as a source of vitamin A. *J. Sci. Food Agric.* 95 (9), 1786–1794. <https://doi.org/10.1002/jsfa.6942>.
- Burri, B.J., La Frano, M.R., Zhu, C., 2016. Absorption, metabolism, and functions of β -cryptoxanthin. *Nutr. Rev.* 74 (2), 69–82. <https://doi.org/10.1093/nutrit/nuv064>.
- Carvalho, N.S., Silva, M.M., Silva, R.O., Nicolau, L.A., Sousa, F.B., Damasceno, S.R., Silva, D.A., Barbosa, A.L., Leite, J.R., Medeiros, J.V., 2015. Gastroprotective properties of cashew gum, a complex heteropolysaccharide of *Anacardium occidentale*, in naproxen-induced gastrointestinal damage in rats. *Drug Dev. Res.* 76 (3), 143–151. <https://doi.org/10.1002/ddr.21250>.
- Chi, T.Y., Zhu, H.M., Zhang, M., 2018. Risk factors associated with nonsteroidal anti-inflammatory drugs (NSAIDs)-induced gastrointestinal bleeding resulting on people over 60 years old in Beijing. *Medicine* 97 (18), e0665. <https://doi.org/10.1097/MD.0000000000001665>.
- Cryer, B., Mahaffey, K.W., 2014. Gastrointestinal ulcers, role of aspirin, and clinical outcomes: pathobiology, diagnosis, and treatment. *J. Multidiscip. Healthc.* 7, 137–146. <https://doi.org/10.2147/JMDH.S54324>.
- Da Silveira Vasconcelos, M., Gomes-Rochette, N.F., de Oliveira, M.L., Nunes-Pinheiro, D. C., Tomé, A.R., Maia de Sousa, F.Y., Pinheiro, F.G., Moura, C.F., Miranda, M.R., Mota, E.F., de Melo, D.F., 2015. Anti-inflammatory and wound healing potential of cashew apple juice (*Anacardium occidentale* L.) in mice. *Exp. Biol. Med.* 240 (12), 1648–1655. <https://doi.org/10.1177/1535370215576299>.
- Das, I., Arora, A., 2017. Post-harvest processing technology for cashew apple – a review. *J. Food Eng.* 194, 87–98. <https://doi.org/10.1016/j.jfoodeng.2016.09.011>.
- da Silva, G.G., Della Torre, A., Braga, L.E.O., Bachiega, P., Tinti, S.V., de Carvalho, J.E., Dionisio, A.P., Ruiz, A.L.T.G., 2020. Yellow-colored extract from cashew byproduct - nonclinical safety assessment. *Regul. Toxicol. Pharmacol.* 115, 104699. <https://doi.org/10.1016/j.yrtph.2020.104699>.
- Diehl, K.H., Hull, R., Morton, D., Pfister, R., Rabemampianina, Y., Smith, D., Vidal, J.M., van de Vorstenbosch, C., 2001. European Federation of Pharmaceutical Industries Association and European Centre for the Validation of Alternative Methods. A good practice guide to the administration of substances and removal of blood, including routes and volumes. *J. Appl. Toxicol.* 21 (1), 15–23. <https://doi.org/10.1002/jat.727>.
- Dionísio, A.P., de Abreu, F.A.P., de Brito, E.S., Wurlitzer, N.J., Ribeiro, P.R.V., de Goes, T. S., Sousa, J.M.S., Iunes, M.F., de Lima, A.C.V., Oiram Filho, F., 2018. Extrato concentrado de carotenóides obtido da fibra do pedúnculo de caju: avaliação da toxicidade com uso do bioensaio de *Artemia salina*. *Boletim de Pesquisa e Desenvolvimento da Embrapa Agroindústria Tropical*. Fortaleza. n. 162 available at: <https://www.embrapa.br/busca-de-publicacoes/-/publicacao/1091917/extrato-co-ncentrado-de-carotenoides-obtido-da-fibra-do-pedunculo-de-caju-avaliacao-da-toxicidade-com-uso-do-bioensaio-de-artemia-salina>. assessed 09/23/2020.
- El-Benna, J., Dang, P.M., Gougerot-Pocidalo, M.A., Elbim, C., 2005. Phagocyte NADPH oxidase: a multicomponent enzyme essential for host defenses. *Arch. Immunol. Ther. Exp.* 53 (3), 199–206. PMID: 15995580.
- Encarnação, S., Mello-Sampayo, C., Nuno, A.G.G., Catarino, L., da Silva, I.B.M., Lima, B. S., Silva, O.M.D., 2016. Total phenolic content, antioxidant activity and pre-clinical safety evaluation of an *Anacardium occidentale* stem bark Portuguese hypoglycemic traditional herbal preparation. *Ind. Crop. Prod.* 82, 171–178. <https://doi.org/10.1016/j.indcrop.2015.11.001>.
- España, I., Jiménez-Moreno, N., Bimbela, F., Ancín-Azpilicueta, C., Gandía, L.M., 2020. Fruit and vegetable waste management: conventional and emerging approaches. *J. Environ. Manag.* 265, 110510. <https://doi.org/10.1016/j.jenvman.2020.110510>.
- Fiedor, J., Burda, K., 2014. Potential role of carotenoids as antioxidants in human health and disease. *Nutrients* 6 (2), 466–488. <https://doi.org/10.3390/nu6020466>.
- Ganesh, S., Kannan, M., Jawaharlal, M., 2015. Cashew industry - an outlook. *Acta Hort.* 1080, 89–95. <https://doi.org/10.17660/ActaHortic.2015.1080.9>.
- Garbayo, J.L.M., Cañada, M.K., Castelló, I.P., Soler, M.T.F., Torres, R.F., Escrivá, M.M., 2017. Cross-sectional analysis of retrospective cases of hospitalisations for gastropathy caused by non-steroidal anti-inflammatory treatment: risk factors and gastroprotection use. *Eur. J. Hosp. Pharm.* 24, 355–360. <https://doi.org/10.1136/ejpharm-2016-001087>.
- Gruszecki, W.I., Strzałka, K., 2005 May 30. Carotenoids as modulators of lipid membrane physical properties. *Biochim Biophys Acta* 1740 (2), 108–115. <https://doi.org/10.1016/j.bbadis.2004.11.015>.
- Hamad, F.B., Mubofu, E.B., 2015. Potential biological applications of bio-based anacardic acids and their derivatives. *Int. J. Mol. Sci.* 16 (4), 8569–8590. <https://doi.org/10.3390/ijms16048569>.
- Hemshekhkar, M., Sebastin Santhosh, M., Kemparaju, K., Girish, K.S., 2012. Emerging roles of anacardic acid and its derivatives: a pharmacological overview. *Basic Clin. Pharmacol. Toxicol.* 110 (2), 122–132. <https://doi.org/10.1111/j.1742-7843.2011.00833.x>.
- Huang, E.S., Strate, L.L., Ho, W.W., Lee, S.S., Chan, A.T., 2011. Long-term use of aspirin and the risk of gastrointestinal bleeding. *Am. J. Med.* 124 (5), 426–433. <https://doi.org/10.1016/j.amjmed.2010.12.022>.
- Jin, M., Otake, M., Okuyama, A., Itoh, S., Otani, S., Odashima, M., Iwabuchi, A., Konishi, N., Wada, I., Pacheco, I., Itoh, H., Tashima, Y., Masamune, O., Watanabe, S., 1999. Association of 72-kDa heat shock protein expression with adaptation to aspirin in rat gastric mucosa. *Dig. Dis. Sci.* 44 (7), 1401–1407. <https://doi.org/10.1023/a:1026603919224>.
- Kauffman, G., 1989. Aspirin-induced gastric mucosal injury: lessons learned from animal models. *Gastroenterology* 96 (2 Pt 2 Suppl. 1), 606–614. [https://doi.org/10.1016/s0016-5085\(89\)80056-3](https://doi.org/10.1016/s0016-5085(89)80056-3).
- Kaulmann, A., Bohn, T., 2014. Carotenoids, inflammation, and oxidative stress implications of cellular signaling pathways and relation to chronic disease prevention. *Nutr. Res.* 34 (11), 907–929. <https://doi.org/10.1016/j.nutres.2014.07.010>.
- Kiela, P.R., Ghishan, F.K., 2016. Physiology of intestinal absorption and secretion. *Best Pract. Res. Clin. Gastroenterol.* 30 (2), 145–159. <https://doi.org/10.1016/j.bpg.2016.02.007>.
- Kim, M.K., Choi, K.Y., Lee, W.C., Park, J.H.Y., Sung, M.-K., Kim, J.-S., et al., 2007. Low serum β -carotene is associated with the incidence of colorectal adenoma. *Nutr. Res.* 27, 127–132. <https://doi.org/10.1016/j.nutres.2007.01.006>.
- Klebanoff, S., 2005. J. Myeloperoxidase: friend and foe. *J. Leukoc. Biol.* 77 (5), 598–625. <https://doi.org/10.1189/jlb.1204697>.
- Krawisz, J.E., Sharon, P., Stenson, W.F., 1984. Quantitative assay for acute intestinal inflammation based on myeloperoxidase activity. Assessment of inflammation in rat and hamster models. *Gastroenterology* 87 (6), 1344–1350. [https://doi.org/10.1016/0016-5085\(84\)90202-6](https://doi.org/10.1016/0016-5085(84)90202-6).
- Kubo, I., Masuoka, N., Ha, T.J., Tsujimoto, K., 2006. Antioxidant activity of anacardic acids. *Food Chem.* 99, 555–562. <https://doi.org/10.1016/j.foodchem.2005.08.023>.
- Lim, T.K., 2012. *Anacardium occidentale*. In: *Edible Medicinal and Non-medicinal Plants*. Springer, Dordrecht, pp. 45–68. https://doi.org/10.1007/978-90-481-8661-7_6.
- Lu, M.S., Fang, Y.J., Chen, Y.M., Luo, W.P., Pan, Z.-Z., Zhong, X., et al., 2015. Higher intake of carotenoid is associated with a lower risk of colorectal cancer in Chinese adults: a case-control study. *Eur. J. Nutr.* 54 (4), 619–628. <https://doi.org/10.1007/s00394-014-0743-7>.
- Malfertheiner, P., Chan, F.K., McColl, K.E., 2009. Peptic ulcer disease. *Lancet* 374 (9699), 1449–1461. [https://doi.org/10.1016/S0140-6736\(09\)60938-7](https://doi.org/10.1016/S0140-6736(09)60938-7).
- Milani, A., Basirnejad, M., Shahbazi, S., Bolhassani, A., 2017. Carotenoids: biochemistry, pharmacology and treatment. *Br. J. Pharmacol.* 174 (11), 1290–1324. <https://doi.org/10.1111/bph.13625>.
- Miller, M.D., Krangel, M.S., 1992. Biology and biochemistry of the chemokines: a family of chemotactic and inflammatory cytokines. *Crit. Rev. Immunol.* 12 (1–2), 17–46. <https://www.ncbi.nlm.nih.gov/pubmed/1418604>.
- Moore, K.W., de Waal Malefyt, R., Coffman, R.L., O'Garra, A., 2001. Interleukin-10 and the interleukin-10 receptor. *Annu. Rev. Immunol.* 19, 683–765. <https://doi.org/10.1146/annurev.immunol.19.1.683>.

- Morais, S.M., Silva, K.A., Araujo, H., et al., 2017. Anacardic acid constituents from cashew nut shell liquid: NMR characterization and the effect of unsaturation on its biological activities. *Pharmaceuticals* 10 (1), 31. <https://doi.org/10.3390/ph10010031>.
- Morais, T.C., Pinto, N.B., Carvalho, K.M., Rios, J.B., Ricardo, N.M., Trevisan, M.T., Rao, V.S., Santos, F.A., 2010. Protective effect of anacardic acids from cashew (*Anacardium occidentale*) on ethanol-induced gastric damage in mice. *Chem. Biol. Interact.* 183 (1), 264–269. <https://doi.org/10.1016/j.cbi.2009.10.008>.
- Mosser, D.M., Zhang, X., 2008. Interleukin-10: new perspectives on an old cytokine. *Immunol. Rev.* 226, 205–218. <https://doi.org/10.1111/j.1600-065X.2008.00706.x>.
- Musumba, C., Pritchard, D.M., Pirmohamed, M., 2009. Review article: cellular and molecular mechanisms of NSAID-induced peptic ulcers. *Aliment. Pharmacol. Ther.* 30 (6), 517–531. <https://doi.org/10.1111/j.1365-2036.2009.04086.x>.
- Nair, A.B., Jacob, S., 2016. A simple practice guide for dose conversion between animals and human. *J. Basic Clin. Pharm.* 7, 27–31. <https://doi.org/10.4103/0976-0105.177703>.
- Nakamura, E., Takahashi, S., Matsui, H., Okabe, S., 1998. Interleukin-1 β inhibits growth factor-stimulated restoration of wounded rat gastric epithelial cell monolayers. *Dig. Dis. Sci.* 43 (3), 476–484. <https://doi.org/10.1023/a:1018890419648>.
- Oliveira, V.H., 2008. Cashew crop. *Rev. Bras. Frutic.* 30, 1–3. <http://www.scielo.br/pd/rbf/v30n1/01.pdf>.
- Parker, H., Winterbourn, C.C., 2013. Reactive oxidants and myeloperoxidase and their involvement in neutrophil extracellular traps. *Front. Immunol.* 3, 424. <https://doi.org/10.3389/fimmu.2012.00424>.
- Perini, R., Fiorucci, S., Wallace, J.L., 2004. Mechanisms of nonsteroidal anti-inflammatory drug-induced gastrointestinal injury and repair: a window of opportunity for cyclooxygenase-inhibiting nitric oxide donors. *Can. J. Gastroenterol.* 18 (4), 229–236. <https://doi.org/10.1155/2004/890585>.
- Persson, C., Sasazuki, S., Inoue, M., Kurahashi, N., Iwasaki, M., Miura, T., Ye, W., Tsugane, S., 2008. For the JPHC Study Group, Plasma levels of carotenoids, retinol and tocopherol and the risk of gastric cancer in Japan: a nested case-control study. *Carcinogenesis* 29 (5), 1042–1048. <https://doi.org/10.1093/carcin/bgn072>.
- Pimple, B.P., Kadam, P.V., Patil, M.J., 2012. Ulcer healing properties of different extracts of *Origanum majorana* in streptozotocin-nicotinamide induced diabetic rats. *Asian Pacific J Trop Dis* 2, 312–318. [https://doi.org/10.1016/S2222-1808\(12\)60068-1](https://doi.org/10.1016/S2222-1808(12)60068-1).
- Rajaratnam, K., Schnoor, M., Richardson, R.M., Rajagopal, S., 2019. How do chemokines navigate neutrophils to the target site: dissecting the structural mechanisms and signaling pathways. *Cell. Signal.* 54, 69–80. <https://doi.org/10.1016/j.cellsig.2018.11.004>.
- Rocha, M.V., Rodrigues, T.H., Melo, V.M., Gonçalves, L.R., de Macedo, G.R., 2011. Cashew apple bagasse as a source of sugars for ethanol production by *Kluyveromyces marxianus* CE025. *J. Ind. Microbiol. Biotechnol.* 38 (8), 1099–1107. <https://doi.org/10.1007/s10295-010-0889-0>.
- Saini, R.K., Nile, S.H., Park, S.W., 2015. Carotenoids from fruits and vegetables: chemistry, analysis, occurrence, bioavailability and biological activities. *Food Res. Int.* 76 (3), 735–750. <https://doi.org/10.1016/j.foodres.2015.07.047>.
- Salehi, B., Gültekin-Özgüven, M., Kirkin, C., et al., 2019. *Anacardium* Plants: chemical, nutritional composition and biotechnological applications. *Biomolecules* 9 (9), 465. <https://doi.org/10.3390/biom9090465>.
- Salehi, B., Gültekin-Özgüven, M., Kirkin, C., et al., 2020. Antioxidant, antimicrobial, and anticancer effects of *Anacardium* Plants: an ethnopharmacological perspective. *Front. Endocrinol.* 11, 295. <https://doi.org/10.3389/fendo.2020.00295>.
- Santos, R.P., Santiago, A.A.X., Gadelha, C.A.A., Cajazeiras, J.B., Cavada, B.S., Martins, J. L., et al., 2007. Production and characterization of the cashew (*Anacardium occidentale* L.) peduncle bagasse ashes. *J. Food Eng.* 79, 1432–1437. <https://doi.org/10.1016/j.jfoodeng.2006.04.040>.
- Sen, S., Chakraborty, R., De, B., Mazumder, J., 2009. Plants and phytochemicals for peptic ulcer: an overview. *Pharmacog. Rev.* 3, 270–279. <https://www.phcogrev.com/sites/default/files/PhcogRev-3-6-270.pdf>.
- Sharifi-Rad, M., Fokou, P.V.T., Sharopov, F., et al., 2018. Antiulcer agents: from plant extracts to phytochemicals in healing promotion. *Molecules* 23 (7), 1751. <https://doi.org/10.3390/molecules23071751>.
- Simões, S., Lopes, R., Campos, M.C.D., Marruz, M.J., da Cruz, M.E.M., Corvo, L., 2019. Animal models of acute gastric mucosal injury: macroscopic and microscopic evaluation. *Animal Model Exp Med* 2, 121–126. <https://doi.org/10.1002/ame2.12060>.
- Sousa, J.M.S., de Abreu, F.A.P., Ruiz, A.L.T.G., Silva, G.S., Machado, S.L., Garcia, C.P., Filho, F.O., Wurlitzer, N.J., Figueiredo, E.A., Magalhães, F.E., Muniz, C.R., Zocolo, G. J., Dionísio, A.P., 2020. Cashew apple (*Anacardium occidentale* L.) extract from a by-product of juice processing: assessment of its toxicity, antiproliferative and antimicrobial activities. *J. Food Sci. Technol.* <https://doi.org/10.1007/s13197-020-04594-0>.
- Sung, B., Pandey, M.K., Ahn, K.S., Yi, T., Chaturvedi, M.M., Liu, M., Aggarwal, B.B., 2008. Anacardic acid (6-nonadecyl salicylic acid), an inhibitor of histone acetyltransferase, suppresses expression of nuclear factor-kappaB-regulated gene products involved in cell survival, proliferation, invasion, and inflammation through inhibition of the inhibitory subunit of nuclear factor-kappaBalpha kinase, leading to potentiation of apoptosis. *Blood* 111 (10), 4880–4891. <https://doi.org/10.1182/blood-2007-10-117994>.
- Takeuchi, K., Amagase, K., 2018. Roles of cyclooxygenase, prostaglandin E2 and EP receptors in mucosal protection and ulcer healing in the gastrointestinal tract. *Curr. Pharmaceut. Des.* 24 (18), 2002–2011. <https://doi.org/10.2174/1381612824666180629111227>.
- Turner, M.D., Nedjai, B., Hurst, T., Pennington, D.J., 2014. Cytokines and chemokines: at the crossroads of cell signalling and inflammatory disease. *Biochim. Biophys. Acta* 1843 (11), 2563–2582. <https://doi.org/10.1016/j.bbamcr.2014.05.014>.
- van Beek, T.A., Montoro, P., 2009. Chemical analysis and quality control of *Ginkgo biloba* leaves, extracts, and phytopharmaceuticals. *J. Chromatogr. A* 1216 (11), 2002–2032. <https://doi.org/10.1016/j.chroma.2009.01.013>.
- Wallace, J.L., Keenan, C.M., Cucala, M., Mugridge, K.G., Parente, L., 1992. Mechanisms underlying the protective effects of interleukin 1 in experimental nonsteroidal anti-inflammatory drug gastropathy. *Gastroenterology* 102, 1176–1185. [https://www.gastrojournal.org/article/0016-5085\(92\)90755-N/pdf](https://www.gastrojournal.org/article/0016-5085(92)90755-N/pdf).
- Wallace, J.L., 2008. Prostaglandins, NSAIDs, and gastric mucosal protection: why doesn't the stomach digest itself? *Physiol. Rev.* 88 (4), 1547–1565. <https://doi.org/10.1152/physrev.00004.2008>.
- Watanabe, T., Higuchi, K., Hamaguchi, M., Shiba, M., Tominaga, K., Fujiwara, Y., Matsumoto, T., Arakawa, T., 2004. Monocyte chemoattractant protein-1 regulates leukocyte recruitment during gastric ulcer recurrence induced by tumor necrosis factor- α . *Am. J. Physiol. Gastrointest. Liver Physiol.* 287 (4), G919–G928. <https://doi.org/10.1152/ajpgi.00372.2003>.
- Watanabe, T., Higuchi, K., Tominaga, K., Fujiwara, Y., Arakawa, T., 2001. Acid regulates in inflammatory response in a rat model of induction of gastric ulcer recurrence by interleukin 1 β . *Gut* 48, 774–781. <https://doi.org/10.1136/gut.48.6.774>.